The set of potential feasible solutions are those where we have b_1 expressed in terms of a_1 as follows:

$$b_1 = \left(\frac{MI}{I^*} - 1\right) \cdot \left(\frac{1}{POV^{a_4} - 1}\right)$$

Using the data in our example we have,

$$b_1 = \left(\frac{25.3}{I^*} - 1\right) \cdot \left(\frac{1}{0.30^{0.5} - 1}\right)$$

Figure 5.5b depicts the combinations of b_1 and a_1 resulting in solutions with MI = 84 bp. But notice that there are several combinations of a_1 and b_1 that are not feasible solutions to the problem. Since we have a constraint on b_1 such that $0 \le b_1 \le 1$ we additionally have $425 < a_1 < 800$. Further, empirical evidence has found $b_1 > 0.70$, thus we have $625 < a_1 < 800$.

Performing these types of sensitivity analyses around the model parameters for those parameters with a known interval will greatly help analysts critique models and results.

Data Definitions

We are now ready to begin testing the I-Star model with actual data. As mentioned in Chapter 4, we will be calibrating our model using market tic data.

Using market data as opposed to actual customer order data will provide three major benefits. First, it provides us with a completely independent data set for estimation and allows us to use the customer order data set as our control group that will be used for comparison purposes. Second, using the market data universe will allow us to eliminate some inherent biases in the data due to potential opportunistic trading where investors trade faster and in larger quantities (shares are added to the order) in times of favorable price momentum, and slower and in smaller quantities (shares or cancelled/opportunity cost) in times of adverse price momentum. Third, this eliminates situations where we (i.e., the broker or vendor) are not provided with the complete order because it is traded over multiple days.

This section describes the data sets used to estimate the market impact parameters. The underlying data used for compiling the data statistics is actual tic data. This is also known as time and sales data, and includes the price of the trade, the number of shares transacted, and date and time of the trade. The data is available via the New York Stock Exchange (e.g., TAQ data for trade and quote data) for all securities traded in the US and/or from various third party data vendors.